

Fangda Dai

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Education

University of Waterloo

Honours Bachelor of Computer Science

Expected Apr 2027

Waterloo, ON

- **Relevant Coursework:** Algorithms, Operating Systems, AI, Computer Vision, Application Development

Work Experience

Software Engineer Intern

May 2026 – Present

RedIron Technologies

Remote

- Built a multi-tenant documentation hosting platform for retail customers using **Astro/Starlight**, **Gitea Actions**, and **Caddy**, deploying Git-managed sites across two **OCI** Oracle Linux hosts
- Automated customer life cycle with a **FastAPI** backend and **web UI**, replacing manual scripts with approvals, status tracking, registry updates, and deployment verification.
- Implemented **versioned releases**, **rollback**, and **health monitoring** with cross-host parity and tenant-isolation checks to catch failed deployments and configuration drift
- Packaged **reusable RPMs** for Gitea workspaces and Oracle Linux setup, reducing environment provisioning from **1–3 days** of manual installation and debugging to **under 10 minutes**

AI Software Developer

Sept 2025 – Dec 2025

Correctional Service Canada

Ottawa, ON

- Developed Python ingestion scripts for **1,000+** law and regulatory PDFs, handling chunking, metadata enrichment, embedding generation, and automated Azure AI Search indexing
- Implemented end-to-end features for an AI document-search application using **Python**, **React**, and **Azure**, taking Azure DevOps tickets through development and release
- Partnered weekly with Microsoft engineers to plan the development-to-production transition and delivered monthly **30-minute** briefings to managers on progress and next steps

AI Application Developer

Jan 2025 – Apr 2025

Immigration, Refugees and Citizenship Canada

Ottawa, ON

- Developed an internal AI web application with a **Python/LangChain backend**, Angular front end, database integration, and RAG-based multi-agent workflows for internal users
- Built **hybrid keyword/vector search** with Azure AI Search and managed vector indexes to improve document retrieval across internal knowledge sources
- Designed a self-reasoning workflow node that filtered approximately **30% of hallucinated responses**, improving reliability before results were returned to internal users

Instructional Support Assistant (C++)

May 2024 – Dec 2024, Jan 2026 – Apr 2026

University of Waterloo

Waterloo, ON

- Created and maintained **Bash/Python scripts** to automatically compile and test C++ submissions for **250+** CS246 students, validating **3,000+** assignment solutions per offering
- Led weekly C++/OOP tutorials and 2-hour exam review sessions for **100+ students**, and coordinated assignment grading with **8** graduate teaching assistants

Project

HelloASL — ASL Learning & Sign-Recognition App

Jan 2026 – Apr 2026

- Co-developed a **Kotlin Multiplatform/Compose** ASL learning app for Android and Desktop, with **Supabase**-backed authentication, lesson data, and user progress/history
- Built the Android **Translate** workflow with ViewModel-driven state, English-to-ASL search/history, **CameraX** recording, lifecycle/error handling, and automated unit/UI tests
- Integrated recorded video with a **FastAPI** inference service via Retrofit, using a **pre-trained VideoMAE model fine-tuned on WLASL100** to return sign predictions and confidence scores to the UI

Skills

Languages: Python, C++, C, Kotlin, TypeScript/JavaScript, SQL, Bash, Java

Frameworks: FastAPI, React, Angular, Compose Multiplatform, LangChain, Astro/Starlight

Platforms/Tools: Linux, Git/Gitea, Docker, Node.js/npm, AWS, Azure, Azure AI Search, Supabase, OCI, CI/CD